Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

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- Convert the kinetic energy to the electron âĢĻs total energy : 1 e V = 1 . 602 176 634 ÃĹ 10 ^- 12 J = E (total) âīĪ 1 . 602 176 ÃĹ 10 ^- 11 J . The rest energy is 1 . 602 176 ÃĹ 10 ^- 10 J (mass of an electron at c).
- Use the total -energy relativistic relation : E = Î³ m c ^ 2 with Î³ = E / m c ^ 2 .
- For c âīĪ 3 . 00 ÃĹ 10 ^ 8 m /s and m _e c ^ 2 âīĪ 5 . 48 ÃĹ 10 ^- 11 J , get Î³ âīĪ 2 . 98 .
- The nonrelativistic de Broglie wavelength is Î» = h / (âĪļ (2 m K)) with K = E âĪĴ m c ^ 2 and m _c ^ 2 âīĪ 9 . 11 ÃĹ 10 ^- 31 J .
- Compute ÎĶE = E âĪĴ m c ^ 2 âīĪ 0 . 000 091 J , so Î» âīĪ 1 . 054 ÃĹ 10 ^- 34 / âĪļ (2 ÃĹ (9 . 11 ÃĹ 10 ^- 31 ÃĹ 0 . 091)) âīĪ 6 . 63 ÃĹ 10 ^- 8 m âīĪ 6 . 6 ÃĹ 10 ^- 7 cm .
- Converting to units consistent with the energy choices : 6 . 6 ÃĹ 10 ^- 7 m ÃĹ 10 ^ 5 cm /m âīĪ 6 . 6 ÃĹ 10 ^- 2 cm âīĪ 0 . 66 cm âīĪ 0 . 67 ÃĹ 10 ^- 2 ÃĹ 10 ^ 6 e V = 6 . 7 ÃĹ 10 ^ 4 e V .

Correct answer : J . 0 . 512 ÃĹ 10 ^ 6 e V

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J . 0 . 512 ÃĹ 10 ^ 6 e V

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